Ch 5.1.4-5: More Cross-Validation

Lecture 14 - CMSE 381

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Fri, Oct 4, 2024

Announcements

Last time:

- k-fold CV

This lecture:

- More k-fold CV
- Bias-Variance Tradeoff
- CV for classification

Announcements:

- Exam 1 grades
- HW #4 Posted.
  - Changed Deadline! Due Wednesday Oct 9.

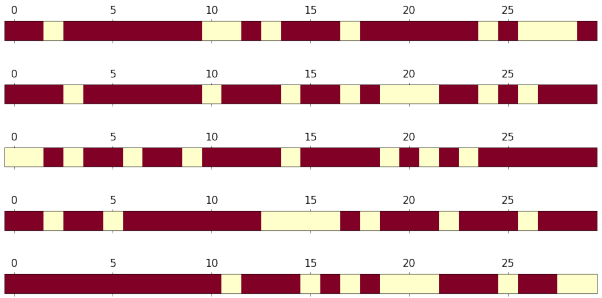
Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

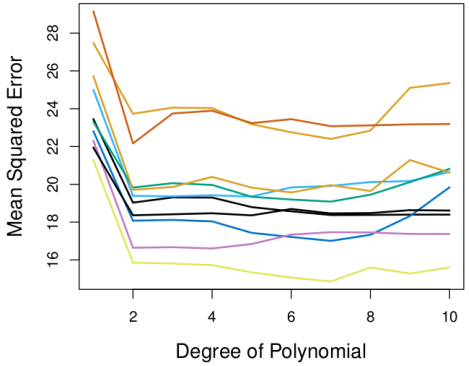
Section 1

k-fold CV

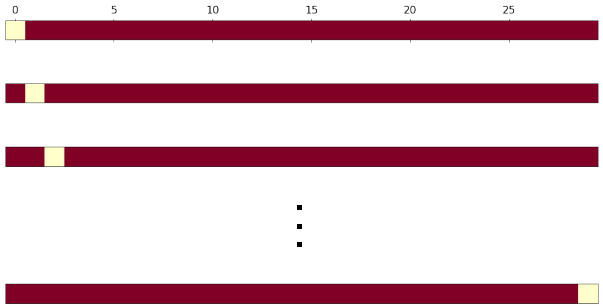
Approximations of Test Error

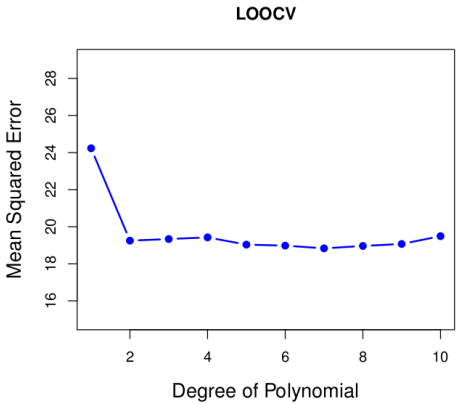
Validation Set



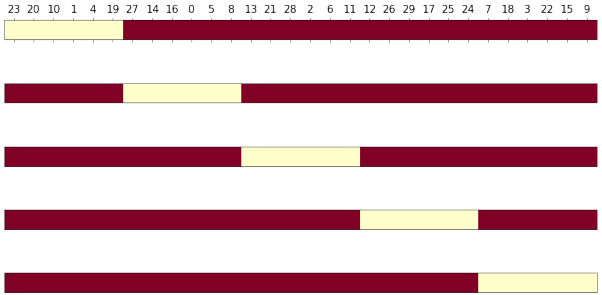


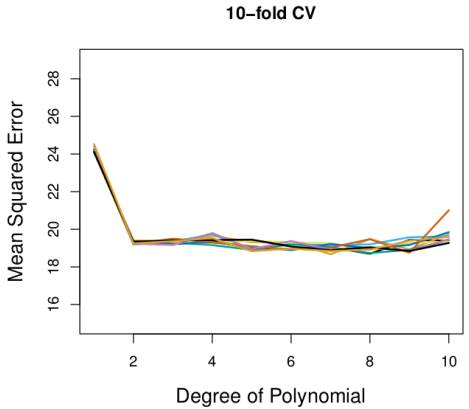
LOOCV





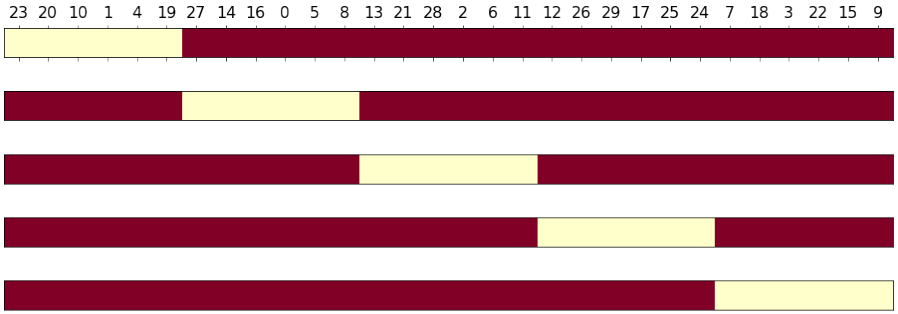
K-fold CV





Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points



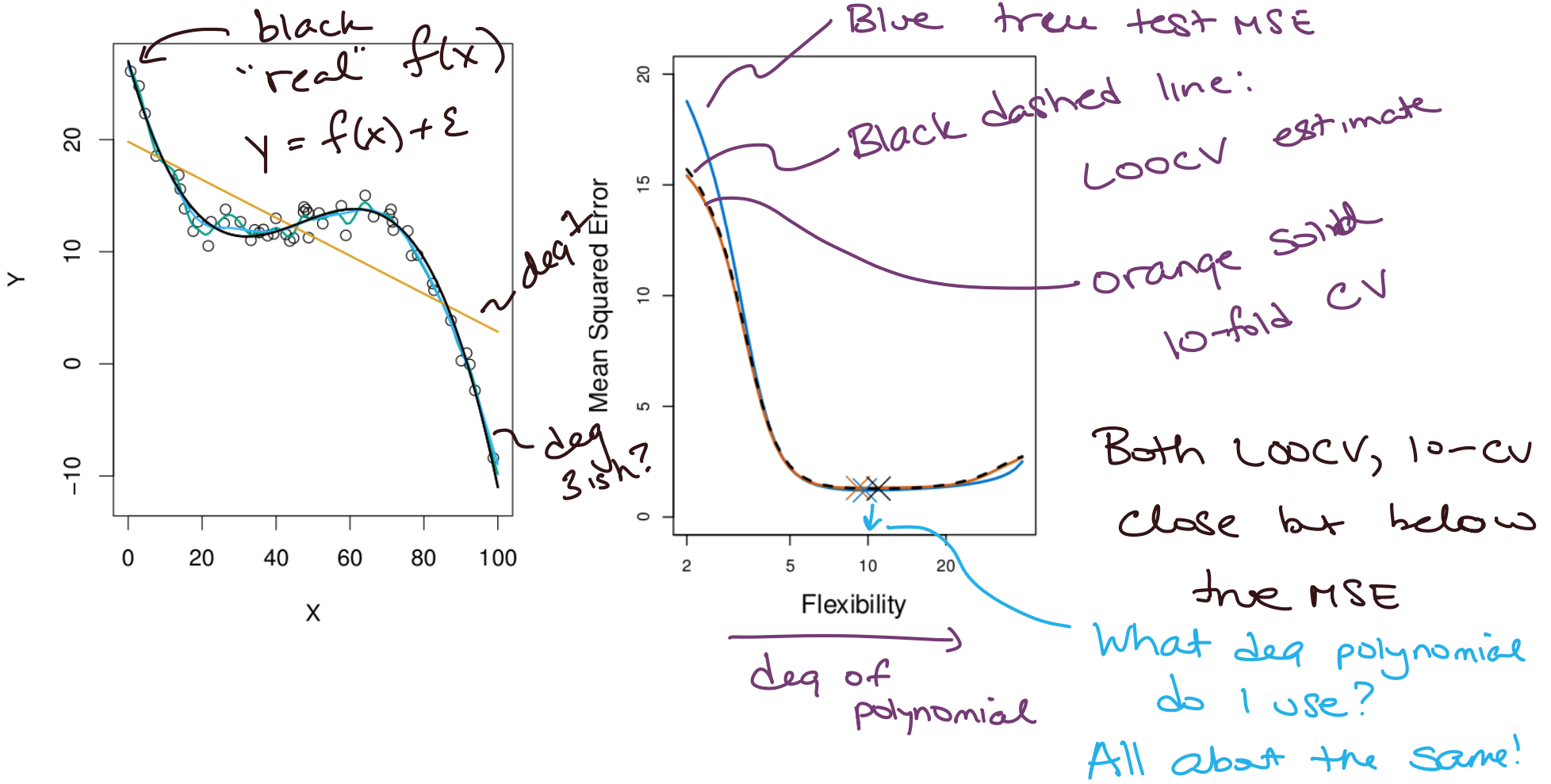
- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$

- Rinse and repeat

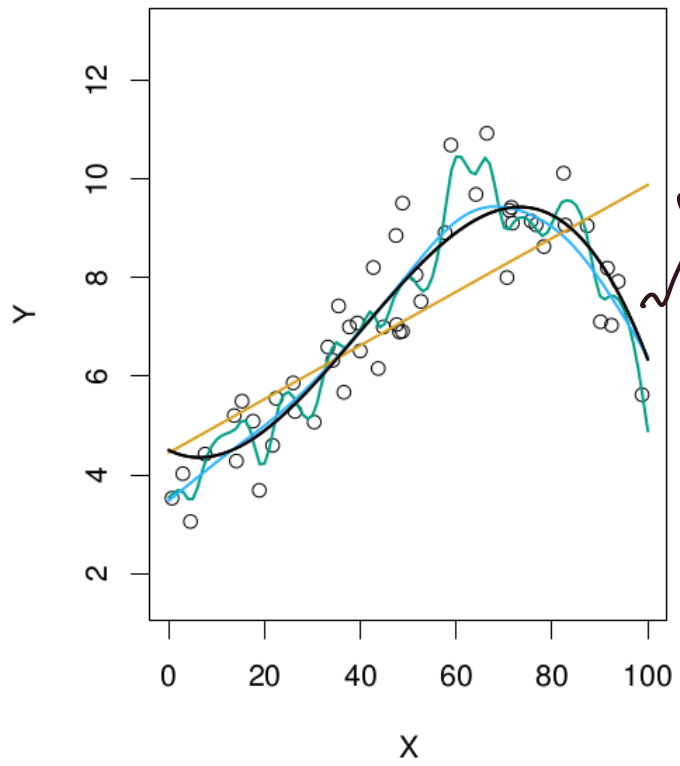
Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

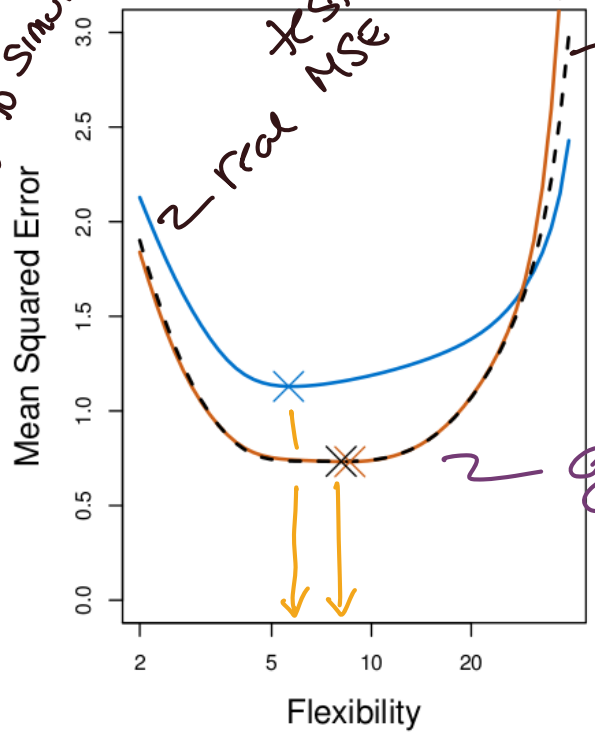
Comparison with simulated data: Ex 3



Comparison with simulated data: Ex 1



f(x) to simulate



test MSE

real

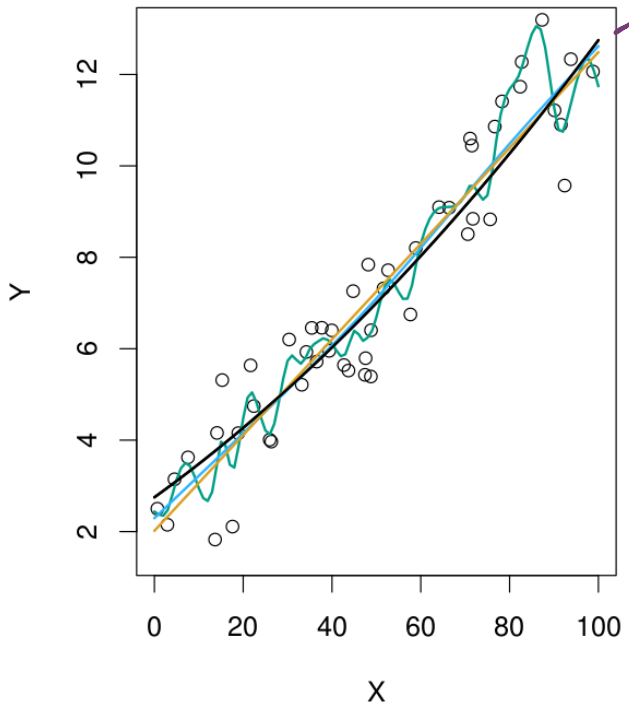
orange 10-fold CV

dashed LOOCV

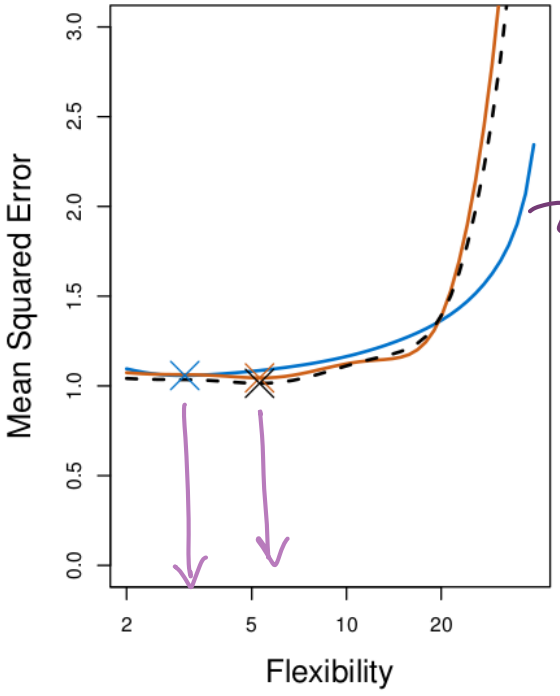
generally under estimating the test error

Same (ish) best model, so its ok that its an under-estimate.

Comparison with simulated data: Ex 2



approx lin data



true test MSE

Takeaways from the examples

- In these examples, LOOCV 10-fold CV are very similar

  Reasonable to use 10-fold CV instead because same results, less computation.

- In the examples, similar x-location for the minimum, so choose similar model whether real MSE or the "approximated" computed version

Bias-Variance Tradeoff: Bias

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

more bias

Validation set tends to over estimate error b/c small subset of data

$(k-1)\frac{n}{k}$ ~ # of pts in fold

k-fold CV medium level of bias b/c training set has approx $\frac{(k-1)n}{k}$ observations

LOOCV gives an approximately unbiased estimate since uses almost all the data all the time

less bias

Bias-Variance Tradeoff: Variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

LOOCV has higher variance (across each MSE_i value)

↳ avg the outputs of n fitted models but all on almost identical observations

↳ high correlation w/ each other.

k-fold

k-fitted models, somewhat less correlated w/ each other

k=5 or 10 is happy medium

mean of many correlated values has higher variance than those not correlated

Section 2

Using K-Fold CV on Polynomial Linear Regression

Polynomial regression

one input variable x_1

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \beta_d x_1^d + \varepsilon_i$$

Can do poly reg. with linear regression by using $(x_1)^\ell$ as the input variables

Faking linear regression into doing our work for us

Input X

$$\begin{bmatrix} 0 \\ 1 \\ 3 \\ 2 \\ -4 \\ -5 \\ \vdots \end{bmatrix} \rightsquigarrow$$

New input matrix X

$$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 \cdots \beta_d x^d$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 3 & 9 & 27 \\ 2 & 4 & 8 \\ -4 & 16 & -64 \\ -5 & 25 & -125 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Coding - Build a plot for train/test scores vs flexibility

Next time

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